(Please write your Exam Roll No.) *Exam Roll No.....*

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

FOURTH SEMESTER - MARCH, 2025

Paper Code: CIC-210 **Subject: Database Management Systems**

Time: 1 ½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No.1 which is compulsory and any two more questions from remaining.

Q. No.	Questions	Max. Marks	CO(s)
Q-1 (a)	We can convert any weak entity set to a strong entity set by simply adding appropriate attributes. Why, then, do we have weak entity sets?	2.5	CO1
(b)	What is data independence in DBMS? Explain logical and physical data independence with the help of a diagram.	2.5	CO1
(c)	Explain the concept of Aggregation in E-R diagrams with the help of an example.	2.5	CO1
(d)	Explain the Division operation in relational algebra with the help of an example.	2.5	CO2
Q-2 (a)	Differentiate between Super key, Candidate key and Primary key with an example.	4	CO1
(b)	Give an example of n-ary relationship (where n=3 and n=4). Also create relations from this relationship.	4	CO1
(c)	How do you map a relationship set to a table in a database?	2	CO1
Q-3(a)	Construct an E-R diagram for a vehicle insurance company whose customers have one or more vehicles. Each vehicle has associated with it zero to any number of recorded accidents.	3	CO1
(b)	Let the following relation schemas be given: R= (A, B, C) S = (D, E, F) Let relations r (R) and s(S) be given. Give an expression in SQL that is equivalent to each of the following queries. (a) $\sigma_{B=17}$ (r) (b) $\Pi_{A,F}$ ($\sigma_{C=D}$ (rxs))	4	CO2
(c)	What is natural join? How do you represent this in relational algebra? Give an example to express this join.	3	CO2
Q-4 (a)	Consider the following relational schema: Sailors (sid, sname, rating, age) Boats (bid, name, color) Reserves (sid, bid, day) Write SQL or Relational algebra expression for each of the following queries: (i) Find the age of the sailors who have ratings of atleast 10. (ii) Find the names of sailors who have reserved boat with id no 27. (iii) Delete all tuples in the boats relation where color of the boat is yellow. (iv) Increase the rating of all sailors by 2%.	8	CO2
(b)	What is the difference between DDL and DML?	2	CO2

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